

Fourat MASTOURI

Availability: ASAP

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SENIOR DATA ENGINEER - 8+ YEARS OF EXPERIENCE

Summary

Freelance Senior Data Engineer with 8+ years of experience across banking, insurance, and healthcare. Expert in data engineering, cloud technologies, and DevOps practices, with extensive experience designing and industrializing complex data pipelines. Technical lead capable of guiding teams, promoting engineering best practices, and adapting quickly to new technical and organizational environments. **Available for new freelance opportunities starting ASAP.**

Key skills: Python (Pandas, Poetry, FastAPI, pytest), PySpark, Microsoft Azure (Databricks, Data Factory, ADLS Gen2), AWS, SQL, Docker, Git.

Professional Experience

AXA France

Nanterre - France

SENIOR DATA ENGINEER - FREELANCE

March 2025 - Present

- As part of the Termostat project, designed and maintained data integration pipelines enabling actuaries to determine appropriate renewal pricing at policy expiry for motor, motorcycle, household multi-risk, and professional multi-risk insurance.
- Delivered reliable, structured datasets supporting actuarial studies and analyses covering quotes, new business, pricing, and portfolio monitoring.
- Provided technical support to actuaries using Databricks environments and conducting analytical work within their labs.
- Held technical ownership of the project's code, artifacts, production environments, and Azure data platform (Azure Databricks, ADF, and ADLS Gen2), ensuring their reliability, maintainability, and smooth operation.
- Contributed to AXA's internal data engineering framework through pull requests, technical improvements, reusable components, and engineering best practices.
- Actively participated in technical chapters by presenting work, proposing cross-cutting improvements, and leading initiatives requiring technical changes across multiple projects.
- Served as a technical lead within the department, guiding teams on architecture decisions, engineering best practices, and complex technical challenges.
- Supported the onboarding of new team members and provided technical coaching and mentoring to interns.

TECHNOLOGIES: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Python, PySpark.

Owkin

Paris - France

SENIOR DATA ENGINEER - FREELANCE

August 2023 - October 2024

- As part of the "DataOps" team at Owkin, contributed to efforts involving the extraction, curation, and alignment of raw medical data, delivering AI-ready datasets for data scientists.
- Established standardized practices for data management, focusing on data quality and delivery for both internal and external projects.
- Engaged in the integration of medical data throughout its lifecycle, including governance, curation, and processing.
- Implemented and maintained CI/CD pipelines to automate testing and deployment, ensuring delivery quality.
- Developed a metadata schema to support various projects.
- Carried out tasks such as extracting, cleaning, curating, pseudonymizing, harmonizing, and merging diverse datasets.
- Provided personalized mentoring and coaching to junior team members to enhance their technical skills and tool proficiency.
- Oversaw junior team members' projects, offering regular guidance and constructive feedback.

TECHNOLOGIES: Python, Poetry, Pydantic, Pandas, AWS, GitHub Actions, DevOps.

Natixis Corporate & Investment Banking

Charenton-le-Pont - France

SENIOR DATA ENGINEER - FREELANCE

February 2022 - June 2023

- Designed, implemented, and maintained data pipelines for the collection, processing, and storage of financial data.
- Developed and optimized ETL processes to ensure data integrity and quality.
- Contributed to the architecture and functional structuring of the data lake.
- Assessed the performance of data processing workflows and identified areas for improvement.
- Worked closely with business teams to understand their data requirements and propose tailored technical solutions to enhance reporting and analysis processes.
- Managed multilingual communication with clients and superiors in English and French.
- Provided guidance and mentorship to junior team members, promoting best practices and knowledge sharing.

TECHNOLOGIES: Python, Spark, Hadoop, XLRise, XLRise, Jenkins, Bitbucket, Control-M, Indexima.

Société Générale

Fontenay-sous-Bois - France

SENIOR DATA ENGINEER - CONSULTANT AT DAVIDSON CONSULTING

March 2019 - February 2022

- Within the Big Data Competency Center (CCBD) and later ITIM/CFT (Fast-IT), contributed to a project aimed at combating internal fraud by industrializing fraud detection workflows in a big data environment.
- Streamlined and operationalized fraud detection processes developed by data scientists.
- Developed data processing and enrichment modules using PySpark.
- Set up data collection pipelines from retail banking data centers to the Hadoop cluster.
- Supported a team of data scientists in leveraging the big data platform effectively.
- Ensured GDPR compliance at the application level.
- Managed communication in English with the support team in Bangalore.
- Handled the operation and maintenance (RUN) of multiple production applications, assisting support teams in incident management and process recovery.

TECHNOLOGIES: Apache Spark, Python, Bash, Oozie, HDFS/Hive, Talend, Control-M.

Paris-Saclay University

Évry - France

ADJUNCT LECTURER - FIXED-TERM CONTRACT

September 2019 - June 2023

- Instructor for the "OLAP & Multidimensional Databases" course in the master's program (M2 IMSD).
- Mentored students and supported some of them as an academic tutor.

Danone Research

Palaiseau - France

DATA ENGINEER - CONSULTANT AT DAVIDSON CONSULTING

September 2018 - February 2019

- As part of the "Biometry & Data Science" team and within Danone's SMART DATA program, my role was to implement data integration workflows.
- Designed, modeled, and implemented data integration processes.
- Conducted a Data Protection Impact Assessment (DPIA) to ensure GDPR compliance.
- Developed a self-service portal enabling non-technical teams to access data from multiple APIs.
- Coordinated with teams in Utrecht and Singapore, acting as a bridge between the two entities to ensure smooth project execution and alignment of the solution with requirements.

TECHNOLOGIES: Cloudera, Apache Spark, Python, Pandas, Alteryx, Impala.

Davidson Consulting

Boulogne Billancourt - France

BIG DATA CONSULTANT - PERMANENT CONTRACT

September 2018 - February 2022

- As a Big Data Consultant, my role was to support Davidson's clients in implementing their data projects within highly distributed environments.
- Contributed to the preparation and drafting of technical sections for responses to requests for proposals (RFPs).
- Assisted Davidson's business engineers in identifying and recruiting new talent while expanding their client portfolios.
- Led workshops to simplify and explain technical concepts to sales teams.

Jems DataFactory / Laboratoire Images, Signaux et Systèmes Intelligents (LISSI)

Neuilly-sur-Seine - France

R&D DATA ENGINEER - APPRENTICESHIP

December 2017 - August 2018

- Implemented various use cases involving Big Data technologies within the JEMS DataLab and the LISSI (Laboratory of Image, Signal, and Intelligent Systems) at UPEC.

TECHNOLOGIES: Hortonworks, ELK Stack, NiFi, Spark, Talend.

Education

Master's Degree

Évry - France

PARIS-SACLAY UNIVERSITY: WORK-STUDY PROGRAM

2018

- Master's degree in IMSD - Innovation, Markets and Data Science.
 - Work-study master's program specializing in Data Science and Data Engineering.
- KEYWORDS:** Big Data technologies, Elasticsearch, Hadoop, Spark, NoSQL, Microsoft Azure, machine learning, data visualization.

Engineering Degree in Information Systems

Tunis - Tunisia

ENICARTHAGE, NATIONAL ENGINEERING SCHOOL OF CARTHAGE

2017

- Engineering school admitting students through the national exam for engineering schools.
- KEYWORDS:** Information systems, development platforms, Java/J2EE, software architecture.

Skills

TECHNOLOGIES:

Data Engineering: Azure Databricks, PySpark, Airflow, Elastic Stack, Kafka, Hadoop ecosystem.
Cloud: Microsoft Azure (Azure Databricks, Azure Data Factory, ADLS Gen2), AWS (EC2, S3).
Programming: Python (FastAPI, Pydantic, Flask, Pandas, Requests), Bash scripting.
Tooling / DevOps: Python (Ruff, uv/Poetry, mypy, PEX, MkDocs, pytest, tox, coverage, setuptools, wheel, pre-commit), Git (rebase, Git Flow, tags, releases, hooks), Make/Just, Docker, GitHub Actions, Jenkins, Ansible.
Databases: PostgreSQL, MySQL, Hive, Indexima, SQLite.

LANGUAGES:

French: Bilingual **English:** Bilingual - TOEIC (985 / 990) $\simeq$ 99%.

Certifications

June 2024 **Astronomer Certification for Apache Airflow Fundamentals**

Astro Academy

Additional Information

- **PREFERRED CONTRACT:** Full-time freelance opportunities only; not interested in permanent roles.
- **REMOTE WORK PREFERENCE:** At least 2.5 remote days per week for Paris-area contracts; negotiable elsewhere.
- **GEOGRAPHICAL MOBILITY:** Available to travel to other regions a few days per month.